

## Training Configuration and Summary:

### Sequencing:

- timesteps: 30
- batch\_size: 128

### Model Hyperparameters:

- min\_units: 50
- max\_units: 150
- units\_step: 50
- dropout\_min: 0.2
- dropout\_max: 0.3
- dropout\_step: 0.1
- activation\_functions: ['tanh']
- optimizers: ['adam']

### Tuning:

- max\_epochs: 20
- hyperband\_iterations: 3

## BiGRU Multi-Step Training Report

Report Generated: 2025-05-03 16:45:23

Training Start: 2025-04-28\_00-13-51

Training End: 2025-04-28\_00-44-38

Training Duration: 30.78 minutes

### Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

MAE: 4.8007

MSE: 24.5554

RMSE: 4.9553

R2 Score: -3.0583

### Multi-Step Forecast:

Forecast Steps: 5

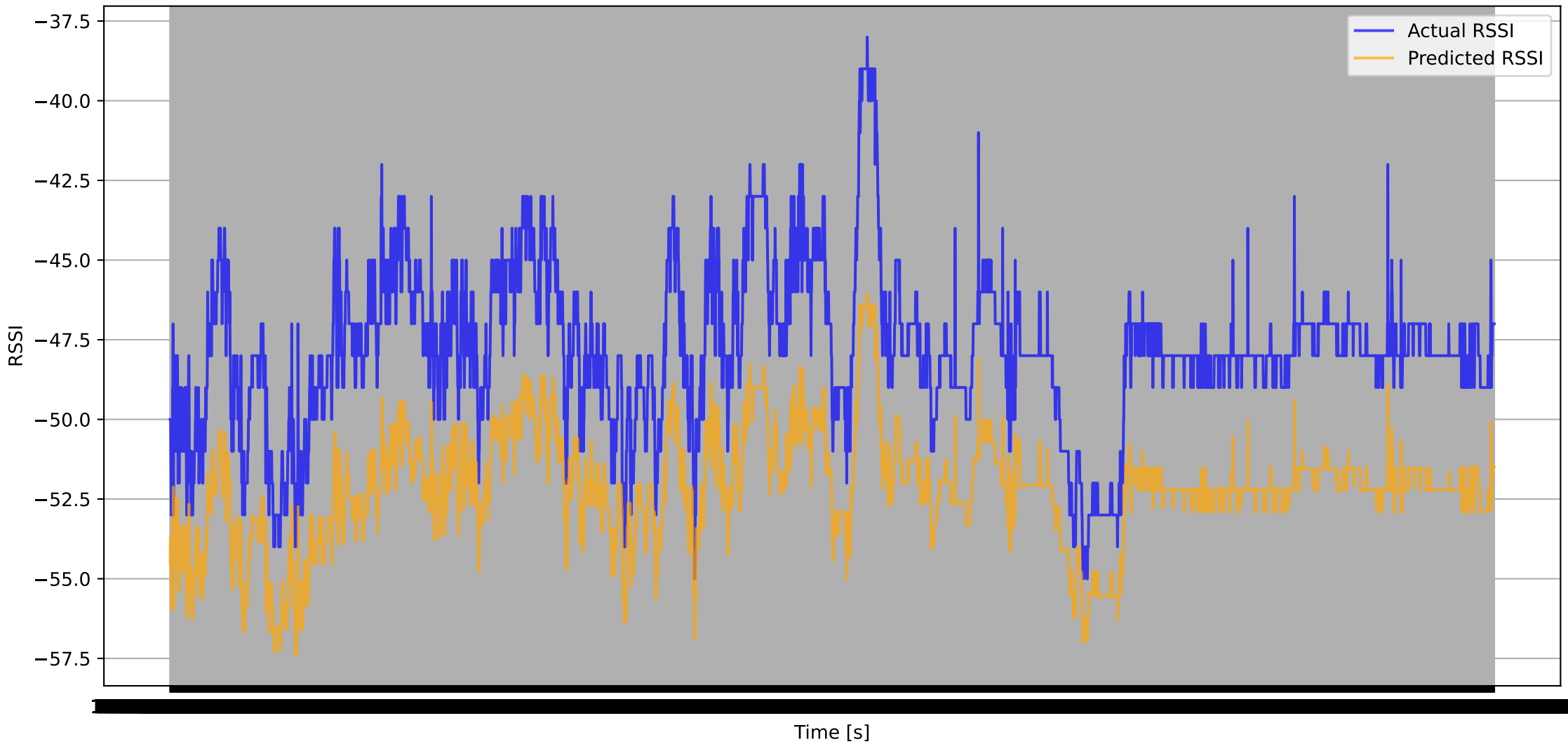
MAE: 0.1508

MSE: 0.0239

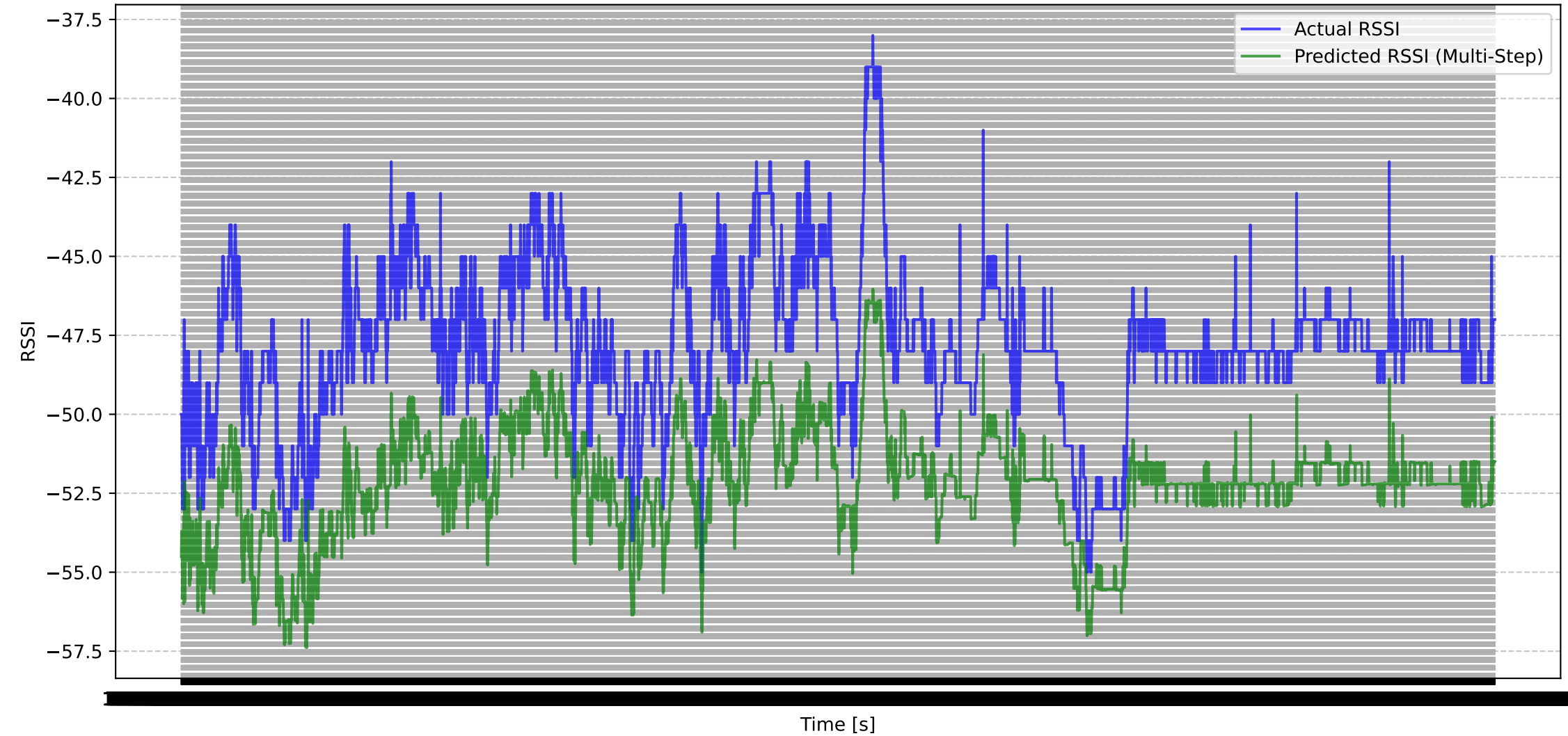
RMSE: 0.1546

R2 Score: -2.3199

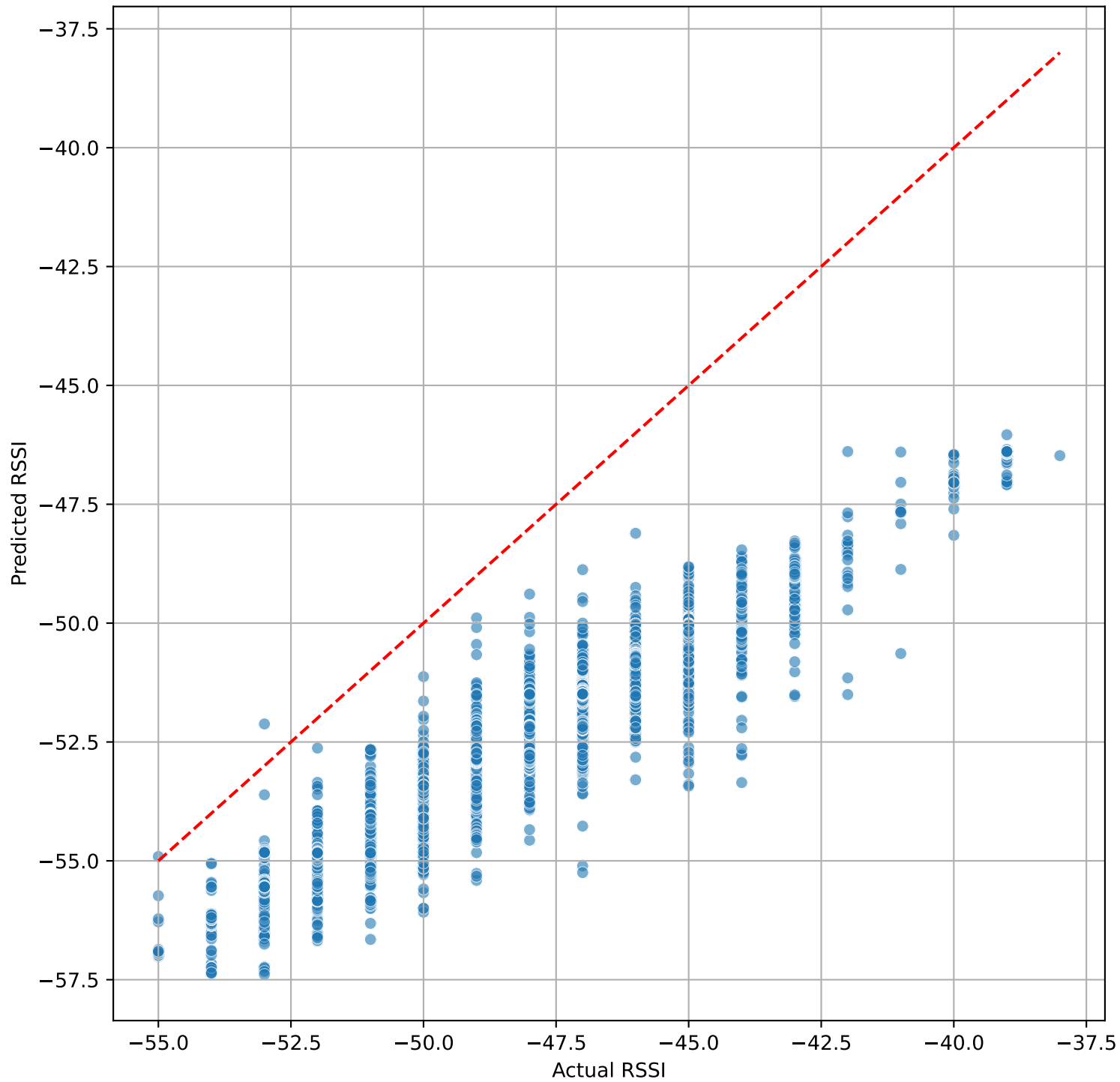
Actual vs Predicted RSSI (First Step)



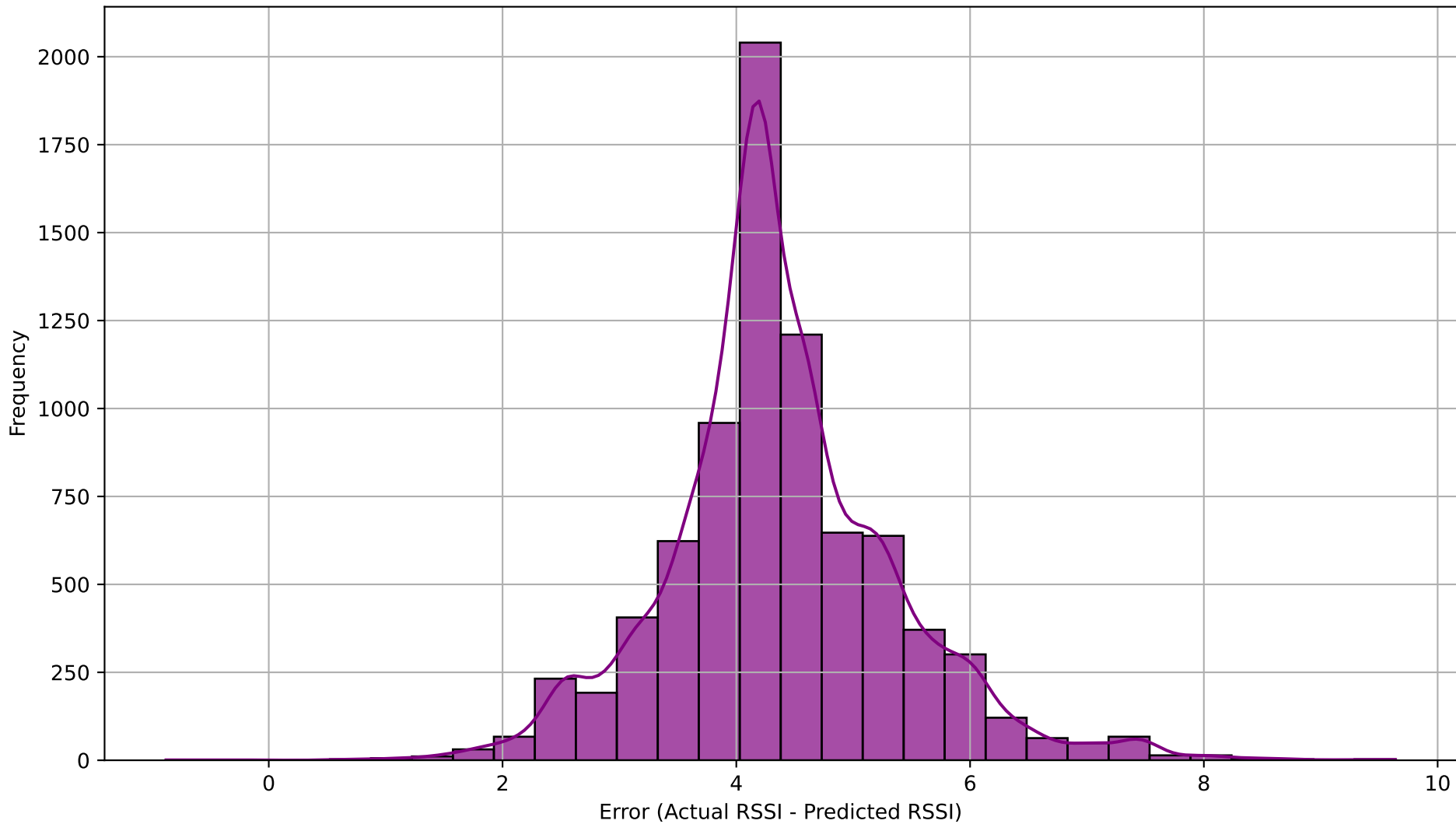
Actual vs Predicted RSSI (First Step vs Multi-Step)



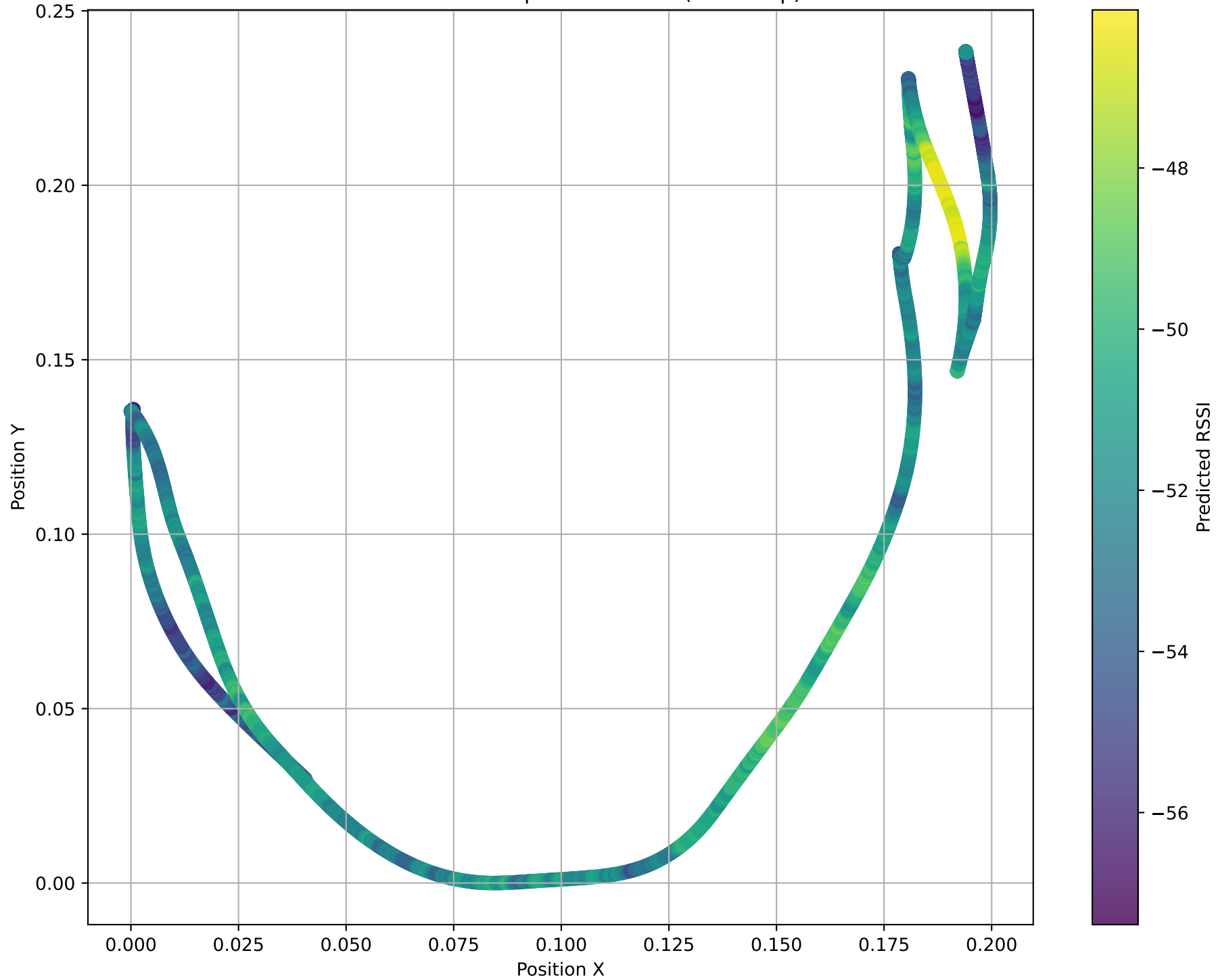
Actual vs Predicted RSSI (First Step)



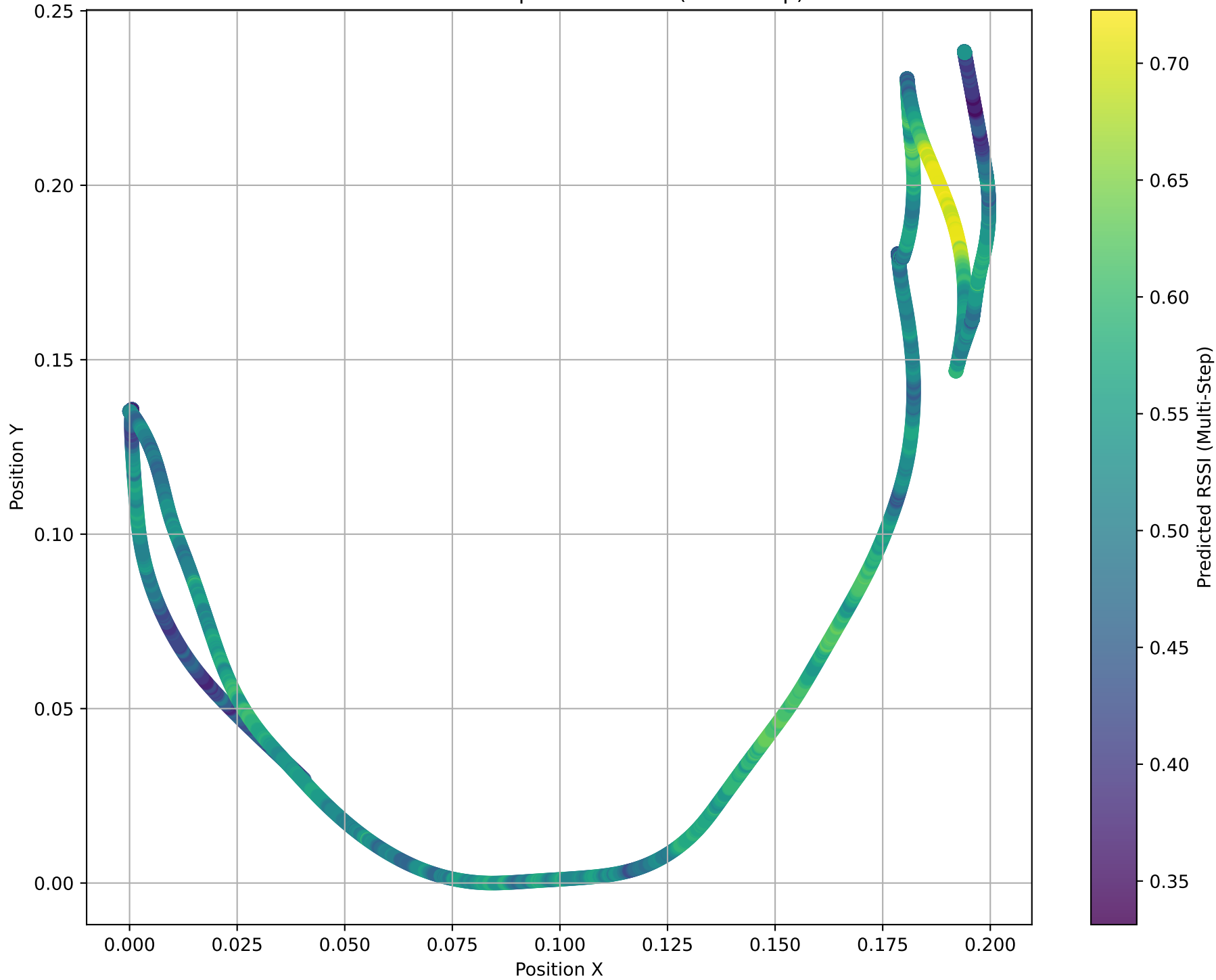
Distribution of Prediction Errors (First Step)



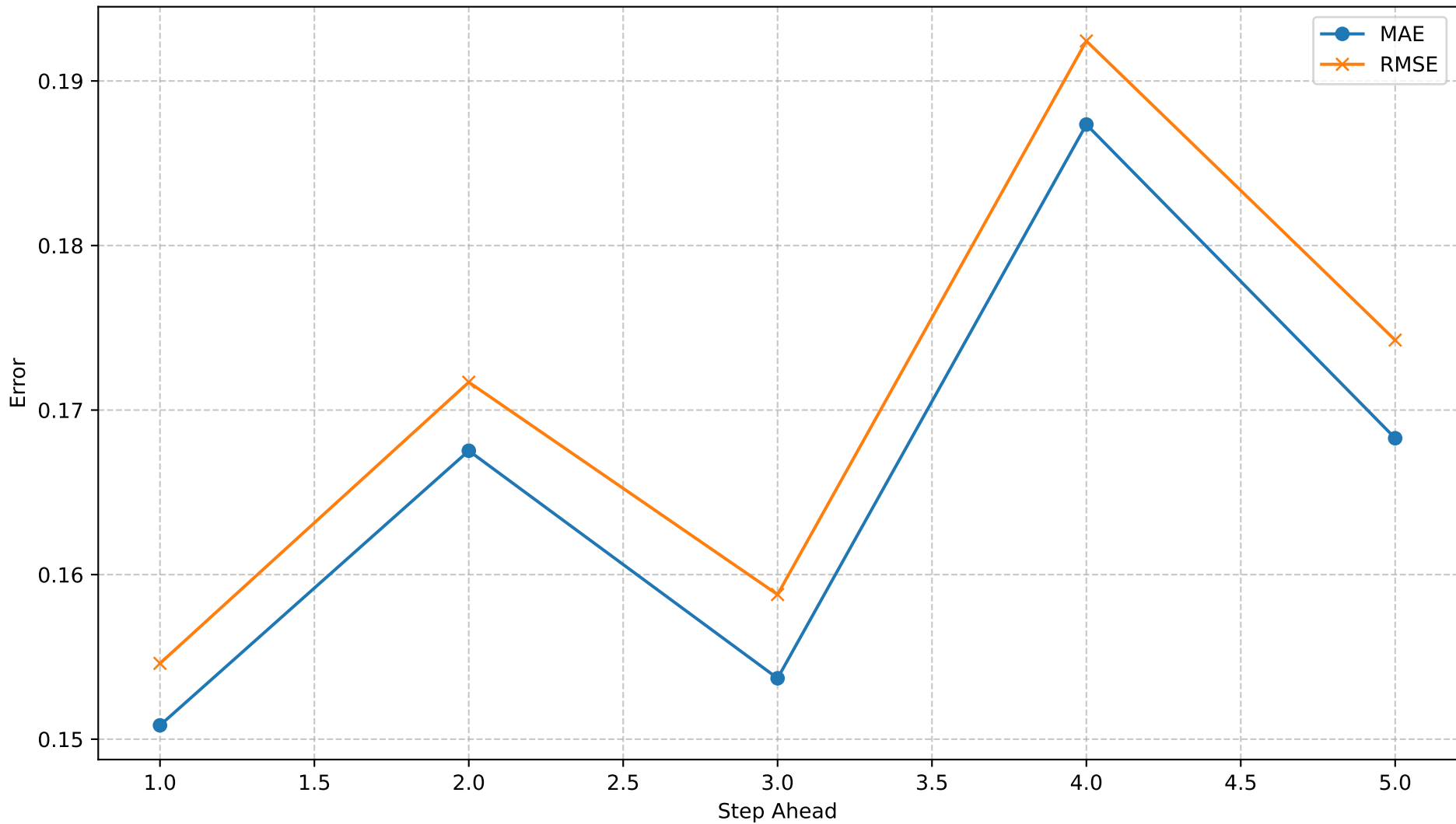
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step)



Multi-Step Forecast Error by Step



#### Resumo Automático:

- Forecast Steps configurados: 5
- Passos com MAE  $\leq 2.0$ : 5 passo(s)
- MAE no último passo: 0.1683

[ V ] O modelo mantém boa acurácia durante quase toda a sequência de previsões.